



SURGIVERSE

VR-Assisted Surgical Simulation & Training Platform

Bridging the global surgical training gap through immersive VR, AI-driven analytics, micro-tactile haptic exoskeleton gloves, and NVIDIA Omniverse-powered robotic AI training.

Unity + OpenXR

NVIDIA Omniverse

dVRK Robotics

Haptic Exoglove

Gemini AI + RAG

Faculty of Engineering, Ain Shams University · Computer & Systems Engineering

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The Problem

Surgical training worldwide is broken.



Ethical & Safety Barrier

"See one, do one, teach one" — trainees learn by risking lives. Practicing on live patients creates genuine ethical dilemmas and patient safety hazards.



Prohibitive Cost

Commercial simulators cost up to \$150,000 (Medtronic Stealth Station S8). Cadaver labs are resource-intensive and largely absent in MENA institutions.



Inequity of Access

Hospitals without morgues — the majority in Egypt and MENA — have no standardized fallback. Trainees in resource-limited settings may never encounter complex cases.



No Measurable Outcomes

Existing training lacks objective, data-driven competency assessment. There is no standardized benchmark to certify surgical readiness.

The Solution

One platform. Every procedure. Zero patient risk.

CT/MRI-Derived 3D Anatomy

- High-res 3D ear models
- CT + endoscope data
- Realistic tissue textures
- Expandable anatomy library

Micro-Tactile ExoSkeleton Glove

- Exoskeleton finger joints
- Micro-actuator tactile buzz
- Force-feedback admittance
- In-house manufactured

AI Performance Analytics

- Gemini AI + RAG scoring
- Real-time error flagging
- Competency certificates
- Progress dashboards

NVIDIA Omniverse Robotic AI

- Digital twin OR environment
- dVRK robotic AI training
- Sim-to-real transfer pipeline
- Surgical dataset generation

Cloud Training Platform

- Firebase auth + Firestore
- Multi-user collab training
- Professor supervision tools
- OpenXR cross-platform

SurgiVerse uniquely combines VR surgical simulation, exoskeleton haptics, AI analytics, and NVIDIA Omniverse robotic training in a single affordable ecosystem — no existing solution integrates all four.

Product & MVP

Ear surgery pilot — production-ready, clinically validated.

3D Ear Anatomy Module

CT-scan + photogrammetry derived high-fidelity model of outer, middle & inner ear. Grommet insertion, tympanoplasty, cholesteatoma removal.

Dexmo Haptic Gloves (v1)

Exoskeleton force-feedback gloves integrated with Unity via custom admittance-control loop. Power system re-engineered for reliability.

Interactive VR Instruments

Endoscope, scissors, and wax-removal tools with collider-based interaction, snapping mechanics, and real-time procedural feedback.

Gemini AI Performance Engine

Firebase + RAG pipeline evaluates trainee actions, flags deviations, and generates PDF competency reports automatically.

TECH STACK

Rendering

Unity 2022 + OpenXR + Meta SDK

3D Modeling

Blender + CT/MRI Photogrammetry

Backend

Firebase Auth + Firestore

AI Layer

Gemini AI + RAG + Cloud Funcs

Haptics

Dexmo SDK + Custom Power System

Robotics

NVIDIA Omniverse + dVRK Isaac

Physics

Kamarianakis 2022 mesh tearing

Deployment

Meta Quest 3 / HTC Vive Pro 2

The Magic — Unique Selling Proposition

No other platform combines all five of these in a single affordable ecosystem.

01

In-House Micro-Tactile Exoskeleton Glove

We design and manufacture our own exoskeleton haptic glove — micro-actuator per finger joint, admittance-control force feedback, off-the-shelf BOM < \$200. No \$15K SenseGlove dependency.

02

NVIDIA Omniverse + dVRK Robotic AI Training

SurgiVerse procedural data feeds an NVIDIA Omniverse digital twin OR. A da Vinci Research Kit (dVRK) robotic arm trains its AI policy on this synthetic surgical dataset — creating a full sim-to-real pipeline.

03

CT-Derived Anatomical Fidelity

High-resolution 3D anatomical models reconstructed from real patient CT/MRI scans and endoscopic footage. Consultant-validated tissue layers, not generic polygon meshes.

04

Embedded AI Competency Engine

Gemini AI + RAG pipeline evaluates every trainee action in real time and benchmarks against expert baselines. Auto-generates certified competency reports — a data-driven alternative to purely subjective assessment.

Business Model

B2B SaaS

Hospital & medical school licensing
\$12K-\$40K/year per institution

Hardware Sales

Exoskeleton haptic gloves
\$450-\$800 per unit (70% margin)

Data Licensing

Anonymized surgical datasets
to robotics & AI labs

Certification API

Per-certificate SaaS fee
\$15-\$50 per competency report

GO-TO-MARKET

Phase 1

Ain Shams University hospital pilot → publish peer-reviewed validation paper

Phase 2

Egypt MoH partnership → onboard 5 public teaching hospitals

Phase 3

MENA expansion

Phase 4

Global licensing + robotic AI dataset monetization pipeline

Market Opportunity



Market Layers



Key Tailwinds

- WHO mandate for competency-based surgical curricula by 2030
- NVIDIA Omniverse opening a \$2B+ surgical robotics AI data market
- Egypt's digital health transformation strategy (Vision 2030)
- Post-COVID acceleration: 74% of residency programs now require simulation exposure

Competitive Landscape

	VR Sim	Real CT Anatomy	Haptic Feedback	AI Scoring	Omniverse Robotics	MENA Focus	Price
Fundamental Surgery	✓	✗	🌕	✗	✗	✗	\$\$\$
Medtronic Stealth S8	✗	✓	✗	✗	✗	✗	\$\$\$\$+
Proximie	🌕	✗	✗	🌕	✗	✗	\$\$\$
Osso VR	✓	✗	✗	🌕	✗	✗	\$\$\$
Surgical Theater	✓	✓	✗	✗	✗	✗	\$\$\$\$
SURGIVERSE ★	✓	✓	✓	✓	✓	✓	\$

No existing competitor combines all six dimensions simultaneously — SurgiVerse's integration is its moat.

The Team



George Joseph Basiliou

Lead Engineer & President HCSD

MLOps · VR Architecture · Embedded Systems · RAG



Youssef Hatem

AI & ML Engineer

Gemini AI · Firebase · RAG Pipeline · Performance Analytics



Michael Shohda

3D Modeling & Unity Dev

CT Reconstruction · Blender · Unity Interactions · OpenXR



Yousif Hazim

Systems & Haptics Engineer

Dexmo Glove Integration · Admittance Control · XR Physics



Youssef Habil

Backend & Cloud Architect

Firebase Auth · Firestore · Cloud Functions · OpenXR API



Mazen Saeed

VR Interaction Developer

Unity XR Toolkit · Collider Systems · Instrument Physics

CLINICAL & ACADEMIC ADVISORS

Prof. Anas Askoura

ENT Prof. & MD, Ain Shams University · CT data provider · Clinical validation

Dr. Nesma Rezk

Asst. Prof. CS&Sys Eng, ASU · Platform Architecture & Technical Supervision

Prof. Tamer Elnady

Prof. Design & Prod. Eng, ASU · Haptic hardware & 3D modeling guidance

Prof. Neveen Asem

VP Innovation, Ain Shams University · Industry partnerships & commercialization

Vision, Next Steps & Timeline

To become the global standard for surgical simulation — replacing cadaveric labs, democratizing training access, and feeding the next generation of autonomous surgical AI.

Sep 2025
– Mar 2026

Phase 1: MVP

- ✓ 3D ear anatomy complete
- ✓ Dexmo haptics integrated
- ✓ Gemini AI scoring live
- ✓ ASU hospital pilot

Mar
– Sep 2026

Phase 2: Validate & Scale

- Peer-reviewed validation paper
- Exoskeleton glove v2 (in-house mfg)
- 5 Egyptian teaching hospitals
- SurgiVerse Technologies startup

Sep 2026
– Mar 2027

Phase 3: Omniverse + Robotics

- NVIDIA Omniverse OR digital twin
- dVRK robotic AI policy training
- Surgical dataset licensing pipeline
- 3 MENA country expansions

2027
– 2029

Phase 4: Global Platform

- 100+ institution licensing
- Certified surgical AI curriculum
- WHO competency-standard adoption
- Series A fundraise

Invest in SurgiVerse — every dollar trains a surgeon without risking a patient.

Investment Required — Budget Breakdown

Pre-seed target: EGP 500,000 – 750,000 · Q3 2026 · Post-grant phase · ~\$10,000–15,000 USD equivalent

Already covered by ASU grant + advisors: Dexmo haptic gloves (Prof. Elnady) · CT/MRI scan data (Prof. Askoura) · ASU lab space & compute

Component	Description	EGP (Low)	EGP (High)	% of Total	Priority
GPU Workstation	High-end GPU workstation (RTX 4080/4090 tier) for Omniverse + Unity rendering	180,000	220,000	33%	CRITICAL
VR Headset (×1)	Meta Quest 3 — import + Egypt customs duties at real market price	45,000	55,000	9%	CRITICAL
DIY Exoglove R&D	PCB design, micro-actuators, 3D-printed exo-frame — 3-unit in-house prototype batch	45,000	65,000	9%	CRITICAL
R&D Stipends (6 engineers)	6-month post-grant stipends at EGP 6,000–8,000/engineer/month (Egyptian market)	216,000	288,000	40%	HIGH
Software Licenses	Firebase Blaze overage, paid Unity plugins, any Omniverse Enterprise licensing	8,000	18,000	2%	HIGH
Clinical Validation	Ethics/IRB approval at ASU + 2–3 pilot hospitals, logistics, travel within Egypt	15,000	25,000	3%	HIGH
Startup Registration	SurgiVerse Technologies Egypt company registration + IP filing fees	5,000	10,000	1%	MEDIUM
Conferences & Publishing	IEEE/ACM submission fees, travel to Cairo-based demo events, printing	10,000	20,000	2%	MEDIUM
Contingency Reserve	Buffer for component failures, import price variance, scope changes	26,000	49,000	6%	BUFFER
TOTAL		EGP 550,000	EGP 750,000	100%	

What We Need From Investors

Capital is one piece — here's the full picture of what accelerates SurgiVerse.

01 Capital — EGP 500K-750K Pre-Seed

- Hardware manufacturing: exoglove pilot batch + GPU workstation
- Clinical validation across 3-5 Egyptian teaching hospitals
- R&D personnel stipends for 6 engineers post-grant period
- Legal startup establishment of SurgiVerse Technologies

02 Hospital & Institutional Access

- Introductions to ENT departments beyond Ain Shams University
- IRB/ethics approval facilitation for clinical pilot studies
- Access to CT/MRI scan repositories for anatomy dataset expansion
- Operating room observation time for surgical workflow validation

03 Industry & Technology Partnerships

- NVIDIA Omniverse academic/startup licensing or grant access
- Huawei Cloud credits for Firebase-equivalent backend at scale
- Hardware supplier introductions for micro-actuator components
- Manufacturing partner for exoskeleton glove production scale-up

04 Strategic Mentorship & Network

- MedTech commercialization guidance (regulatory pathways in Egypt/MENA)
- Connections to MENA digital health accelerators and impact funds
- Academic publishing support for peer-reviewed validation paper
- Advisor introductions: surgical robotics AI and haptics domain experts

The right investor doesn't just write a cheque — they open an operating room door.

References & Resources

Source documents, technology documentation, and clinical evidence base.

Source Documents & Clinical Evidence

SurgiVerse GP Report (ASU 2026)

[uploaded: GP.pdf]

ASU Funded Project Proposal

[uploaded: SurgiVerse_.docx]

Dr. Anas Askoura — VR Surgical Training Deck

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Seymour NE — VR to OR meta-analysis

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Graafland M et al. — Serious games surgery

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Comput Graph 2022 doi:10.1016/j.cag.2022.07.015

Li T et al. — VR orthopedic education meta-analysis

J Med Internet Res 2025;27:e70266

NVIDIA Omniverse & Isaac

NVIDIA Omniverse overview

nvidia.com/en-us/omniverse

Omniverse developer portal

developer.nvidia.com/omniverse

Omniverse workflows docs

docs.omniverse.nvidia.com/workflows/latest

Omniverse Kit SDK

docs.omniverse.nvidia.com/kit/docs/kit-sdk

OpenUSD documentation

openusd.org/docs

Isaac Sim (robot simulation)

developer.nvidia.com/isaac/sim

Isaac Lab (robot learning)

github.com/isaac-sim/IsaacLab

Isaac for Healthcare (i4h)

isaac-for-healthcare.github.io/i4h-docs

Isaac i4h HuggingFace blog

huggingface.co/blog/nvidia/nvidia-isaac-for-healthcare

ORBIT-Surgical (dVRK training)

arxiv.org/pdf/2404.16027

Sim-to-Real bridging (Isaac Lab)

developer.nvidia.com/blog/bridging-sim-to-real-gap

dVRK, Haptics & Platform Stack

dVRK official site (JHU)

dvrk.lcsr.jhu.edu

Intuitive Foundation dVRK apply

intuitive-foundation.org/dvrk

dVRK GitHub org (JHU)

github.com/jhu-dvrk

dVRK ReadTheDocs

dvrk.readthedocs.io

Dexmo (Dexta Robotics) glove

dextarobotics.com

Unity XR Interaction Toolkit

docs.unity3d.com/Packages/com.unity.xr.interaction.toolkit

OpenXR specification (Khronos)

registry.khronos.org/OpenXR/specs/1.1/html/xrspec.html

Firebase docs

firebase.google.com/docs

Gemini AI API docs

ai.google.dev/gemini-api/docs

Vertex AI RAG overview

cloud.google.com/vertex-ai/generative-ai/docs/rag-overview

ASU Innovation Sector

med.asu.edu.eg / eng.asu.edu.eg